

SHAIFALI VASHISTHA

[Mail](#) ◇ [LinkedIn](#) ◇ [Github](#) ◇ [Webpage](#)[†]

EDUCATION

Indian Institute of Technology Madras, Chennai, India

Dec '20 - Present

B.S. in Data Science & Its Applications

GPA: 7.83

Central Board of Secondary Education, Delhi, India

CBSE Intermediate, Percentage: 80.4%

2017-2018

CBSE Matriculation, CPI: 10/10

2015-2016

KEY PROJECTS

Automatic Text Summarizer | Natural Language Processing

Spring '23 - Present

- Implementing an **interactive** text summarizer that automatically produces a summary of a document
- Integrating an approximate summarization model to the original model to guarantee interactive speeds for large text collections of multiple topic-related documents to keep the user engaged in the process

Eigen Face Recognition System | Image Processing

Fall '22

- Implemented quick face recognition & reconstruction using **eigen-faces** obtained from **PCA** & **SVD**
- Represented facial images from the **ORL Dataset** as n -dimensional vectors and implemented dimensionality reduction by obtaining their eigenvectors acting as key signatures in the recognition task

Mentorship platform Application | Software Engineering

Fall '22

- Created a mentorship platform for IIT Madras BS students using **Flask** & **RESTful API** for back-end, **Jinja 2** with **HTML**, **CSS** & **bootstrap** for user interface based on **Agile** approach
- Collected user requirements as user stories through online interviews, designed the interactive wireframe of the application using **figma** and used **Jira** Software for keeping track of sprint meetings

KanBan ToDo Application | Application Development

Fall '22

- Developed a python web application using **Flask-SQLAlchemy** and **SQLite3** for managing and visualizing tasks, setting deadlines with daily task reminder emails and monthly statistical reports
- Used Flask framework in python for back-end development, **Vue-CLI** for front-end development and **Celery** & **Redis** for **caching** & implementation of triggered and scheduled tasks in the application

Financial Position Analysis | Business Data Management

Fall '21

- Analyzed the **revenue trend** and **net profit trend** over the last 5 years for the State Bank of India
- Conducted numerical analysis of key **financial ratios** & derived insights on the financial performance
- Compared the financial performance of the bank to its peers and documented the results in a report

Model Training from Scratch | Machine Learning

Spring '22

- Implemented machine learning algorithms such as **Logistic Regression**, **Gaussian Naive Bayes**, **Softmax Regression**, **KNN**, Soft Margin **SVM** and **Kernel SVM**, and **ANN** to train models

SCHOLASTIC ACHIEVEMENTS

- Completed **Degree in Data Science & Its Applications, 2022** from **IIT Madras**
- Awarded **S grade** in Mathematics, Programming in C, Modern Application Development, and Business Analytics
- Achieved **97.8** percentile in **AIIMS-MBBS 2018** among 374,000 candidates
- Secured **90** percentile in **NEET (UG) 2018** among 1.59 million candidates

[†]Use URL www.shailivashistha.github.io in case hyperlinks don't work

POSITIONS OF RESPONSIBILITY

Teaching Assistant | *Modern Application Development*

Spring '23

- Will be responsible for organizing regular doubts solving sessions, handling course-related queries, conducting programming sessions, and preparing mock tests & solutions for a batch of **40 students**

Mentor | *Modern Application Development*

Fall '22

- Organised regular doubt-clearing sessions and managed weekly assignments for a batch of 21 students

OTHER PROJECTS

Mask Data Analysis | *Data Visualization and Design*

Spring '23

- Created a web application to log and manage boolean and numeric data for daily activities using **Flask** with **Jinja2** templates for backend and **HTML5, CSS & Bootstrap** for frontend development
- Modified Self Tracker from a **static** web application to an advanced **dynamic** web application using **Vue-CLI** for frontend development, and **flask** for backend development of the application.

Self Tracker Application | *Application Development*

Summer '22

- Created a web application to log and manage boolean and numeric data for daily activities using **Flask** with **Jinja2** templates for backend and **HTML5, CSS & Bootstrap** for frontend development
- Modified Self Tracker from a **static** web application to an advanced **dynamic** web application using **Vue-CLI** for frontend development, and **flask** for backend development of the application.

Predictions from Data using sklearn | *Machine Learning*

Spring '22

- Designed and trained various models using **sklearn** to make predictions over data and detect outliers

E-Commerce Trend Analysis | *Business Data Management*

Fall '21

- Used a dummy data corresponding to an E-Commerce company for **trend analysis** of the firm
- Analysed distribution of revenue, stock price, and volume trends using **Pareto** and **scatter plots**
- Utilised Google sheets to present various trends in line charts, bar charts, and combination charts

TECHNICAL SKILLS

Programming Python, C, C++, SQL, JAVA, Vuejs, MATLAB, Octave, bash

Libraries Pytorch, NumPy, Pandas, SciPy, SKLearn, Flask, BeautifulSoup, tabula

Softwares Linux, Github, PostgreSQL, MS Excel, SQLite, QGIS, Redis, Jira, Figma, L^AT_EX

RELEVANT COURSES

ML & AI NLP (*NPTEL*), Search Methods for Problem-Solving (AI), Reinforcement Learning, Deep Learning, Machine Learning Techniques, Machine Learning Practice, ML Foundations

CSE Software Testing, Ethical Hacking (*NPTEL*), Software Engineering, Cloud Computing (*NPTEL*), Data Structures & Algorithms, Advanced Algorithms, Programming in C, Computer System Design, Database Management Systems, Programming in Java, System Commands, Advanced Application Development

Other Courses Discrete Mathematics (*NPTEL*), Market Research, Industry 4.0, Data Visualisation & Design, Tools in Data Science, Business Data Management, Business Analytics, Managerial Economics, Mathematics I & II, Statistics I & II

EXTRACURRICULAR ACTIVITIES

- Participated in **Datathon** hosted by **IBM Zurich** and **Shooting Stars Foundation** (2022)
- Participated in **Capture the Flag** contest at **Shaastra**, organized by IIT Madras (2022)
- Participated in **Reverse coding contest** at **Shaastra**, organized by IIT Madras (2022)